

Tarang Singh

Department of Physics, Indian Institute of Technology (IIT-BHU)



PROFESSIONAL EXPERIENCE

Tata Institute of Fundamental Research

Winter Project, Dr. Mandar Deshmukh

Colaba, Mumbai

October 2022 - November 2022

- Exfoliating WSe₂ and h-BN to study the 2-dimensional properties
- Trained in nanoelectronics lab and clean room

National Initiative for Undergraduate Science 19.1, TIFR-HBCSE

Trombay, Mumbai

December 2022- January 2023

- Attending a series of lectures and experiments designed by scientist from various top research institutes in India designed to give undergraduate students a better exposure in research conducted in physical sciences
- Selected as one of 60 students from all over India to attend the initiative

National Initiative for Undergraduate Science 19.2, TIFR-HBCSE

Trombay, Mumbai

Research Project, Dr. Sudhir Jain

2023

- Selected through a test conducted for the 60 students selected through the initiative
- Understood the aspects of Quantum error corrections, basics of Quantum Computing and Genus Codes.

Summer Research Scholar

Dr. Rajeev Singh

IIT-BHU, Varanasi

May 2026 - July 2026

- First Author Paper (currently arxivd and undergoing reviews, to be published) on tuning magnetic filling factor for spiral microwave resonator coupling with YIG creating a Hybrid Quantum System
- Non-monotonic coupling dependence on turn number, $g/2\pi \approx 50-205$ MHz, characterization across $N = 1, 3, 5, 7, 9$
- Experiment using Virtual Network Analyser (VNA), and performing to exfoliating techniques and microwave engineering for cavity magnonics

Tutor, Inkstall Educare Pvt Ltd.

Western Suburbs, Mumbai

June 2024-Ongoing

- Teaching Math and Physics to students from Grade VIII to XII, across IGCSE, IBDP, ICSE and CBSE curriculum

EDUCATION

Podar International School, ICSE (90.66%)

Primary and Secondary Schooling

Kalyan, Thane

2008-2019

B.K Birla College, HSC (88.5%)

Higher Secondary Schooling

Kalyan, Thane

2019-2021

D.G Ruparel College (8.9 CGPA)

Bachelors in Science

Matunga, Mumbai

2021-2024

Indian Institute of Technology (IIT-BHU) (ongoing)

Masters in Science

Varanasi, Uttar Pradesh

2025-2027

ACHIEVEMENTS

- Selected for ICTS-TIFR interview stage (**top 40-50** candidates in India through TIFR-GS written test)
- Qualified for TIFR written test (**top 123** in TIFR-GS, 2025)
- All India Rank **343**, JEST-2024
- All India Rank **437**, JAM-2025
- Class Representative, MSc (Batch of 2025), IIT-BHU
- Bronze Medal: Undergraduate Math Poster Competition, D.G. Ruparel College
- Bronze Medal: Undergraduate Intercollege Physics Quiz Competition, K.C. College, Mumbai

PARTICIPATIONS

- VIBGYOR 2023: Built a model demonstrating the effect of body's shape on motion in fluids (viscous forces) and constructed a rocket launcher capable of reaching the second floor of a standard building
- AVISHKAR 2022: Designed and built a Van de Graaff generator from inexpensive materials, achieving a potential difference of approximately 3 kilovolts
- Basic Astronomy Course: Completed at D.G. Ruparel College.
- Advanced Astronomy Course: Completed at D.G. Ruparel College
- Published Article: Featured in "Ganitananda", a mathematics journal by D.G. Ruparel College, with an article titled "Trigonometric Identities: and where did they come from?"
- IAPT Summer School 2022: Attended a summer school focused on problem-solving, organized by the Indian Association of Physics Teachers

PROFICIENCIES

- Lab techniques involving thin-film etching, using a Virtual Network Analyzer, and fabrication of microwave resonators
- Software proficiency in Quantum Metal, MATLAB, and LaTeX
- Exceptional problem-solving abilities and theoretical rigor